EDGE DEVICE SPECIFICATION

MODEL: ProAI77



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Revision History

VERSION	DATE	BOARD ID	DESCRIPTION	AUTHOR
1.0	2026/04/01	ProAiBox77	Specification	Colin
1.1	2026/05/01	ProAIBox77	Update specification	Colin
1.2	2026/05/28	ProAIBox77	Update specification	Colin

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1. General Description

The proAI77 is an edge device with high performance ARM SoC with AI accelerator module inside to provide the best performance for edge computing application.

This product is a highly integrated edge AI computing platform, featuring the ARM-powered single-board computer (based on the Rockchip RK3588 architecture) as the primary control core, equipped with an edge AI accelerator module (Provide around 24 FTOPS) via an M.2 expansion slot. This hardware configuration combines a multi-core CPU/GPU processing unit with a low-latency AI dataflow inference architecture. It effectively offloads computer vision (CV) computing workloads from the main processor, providing an easy-to-integrate AI edge computing solution for commercial, retail, and security surveillance environments.

The following are the main features of proAI88/88J:

- Supports the LPDDR up to 32GB, with extends the EMMC up to 128GB.
- Supports for HDMI output.
- Supports a variety of USB2.0/USB3.0/Type-C peripheral interfaces.
- Supports dual-gigabit cable network port
- Supports multiple operating systems: Debian11 and Ubuntu, Android is optional.
- Supports for GPIO / UART / SPI / I2C / RS232 and other extended interfaces.

For the AI accelerator module, the feature of the AI module provides below features:

- AI frameworks: ONNX, PyTorch, TensorFlow, Keras, TensorFlow Lite
- Weight parameters: 80 million, 4-bit. Or 40 million, 8-bit
- OS Support: Ubuntu 22.04

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Product Pictures



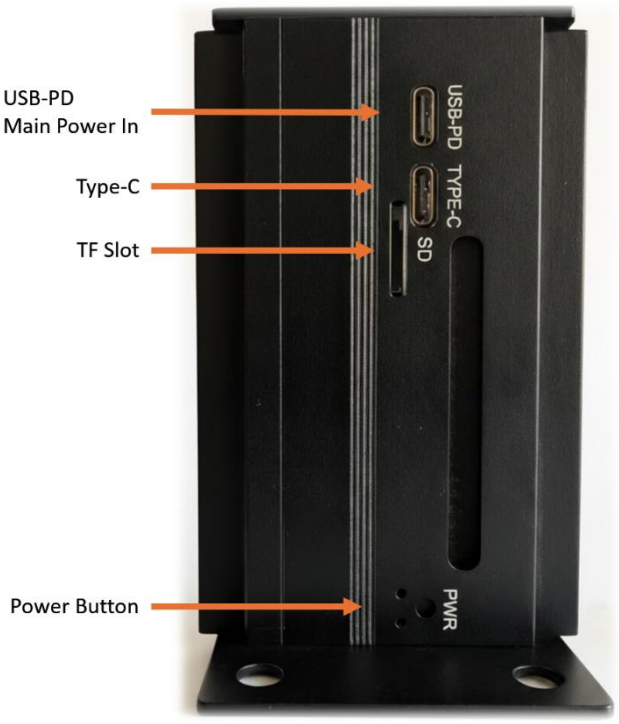
Side View 1

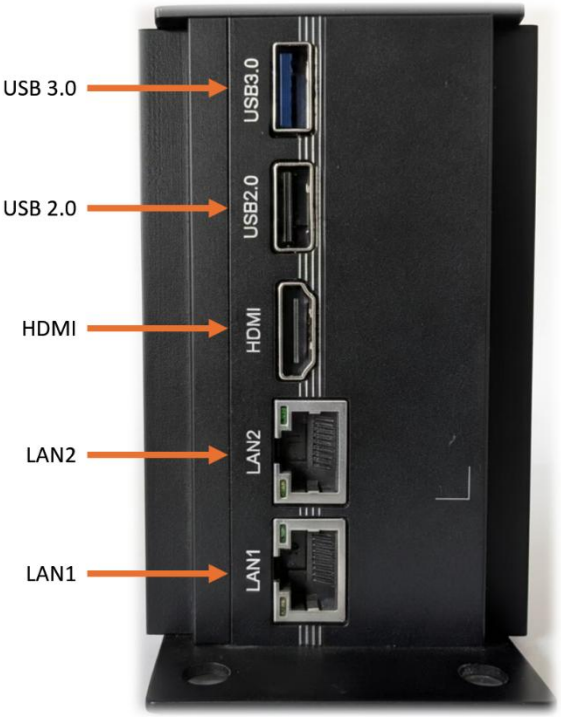


Side View 2

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2. Product Interface





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3. Product Parameter

Market	Commercial, Retail, Machine Vision	
System Language	English. Extend to selected language by optional	
Configure	Operating System	Ubuntu 22.04, Armbian
	CPU	4x Cortex-A76 @ 2.4GHz + 4x Cortex-A55 @ 1.8GHz
	GPU	Mali-G610 MP4
	DDR	8GB (16/32G optional)
	Storage	32GB (64G/128G/256G optional)
Network	Ethernet LAN	2x 2.5G Ethernet
Display	HDMI Out	1 port
GPIO	Extension I/O	By configuration
AI Accelerator Module	Up to 24 TFLOPS (1GHz), 42M 8-bit parameters (weights)	
USB	1x USB 2.0, 1 x USB 3.0, 1x Type-C	
Size	63mm(W) x 100mm(L) x 100.1mm(H)	
Power Pack	USB Type-C PD, 12V/3A	

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4. Supported Formats

Video Decoder

- Real-time video decoder of MPEG-1, MPEG-2, MPEG-4, H.263, H.264, H.265, VC-1, VP9, VP8, MVC, AV1
- MMU Embedded
- Multi-channel decoder in parallel for less resolution
- H.264 AVC/MVC Main10 L6.0 :8K@30fps (7680x4320)
- VP9 Profile0/2 L6.1 :8K@60fps (7680x4320)
- H.265 HEVC/MVC Main10 L6.1 :8K@60fps (7680x4320)
- AVS2 Profile0/2 L10.2.6 :8K@60fps (7680x4320)
- AV1 Main Profile 8/10bit L5.3 :4K@60fps (3840x2160)
- MPEG-2 up to MP :1080p@60fps (1920x1088)
- MPEG-1 up to MP :1080p@60fps (1920x1088)
- VC-1 up to AP level 3 :1080p@60fps (1920x1088)
- VP8 version2 :1080p@60fps (1920x1088)

Video Encoder

- Real-time H.265/H.264 video encoding
- Support up to 8K@30fps
- Multi-channel encoder in parallel for less resolution

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5. Precautions For Use

1. Relative humidity: 10% ~ 90%
2. Operating temperature: 0℃ ~ 80℃
3. Storage temperature: -40℃ ~ 105℃
4. Do not squeeze, distort or disassemble the board.
5. Keep the board away from static electricity.
6. Keep the board away from water and other liquid.
7. Clean the board with soft and clean dry cloth when it’s dirty.
8. Don’t use long connect wires which may affect performance and image quality.

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